Bayesian Inference for the German Tank Problem: Priors, Structured Serial Numbers, and Simulation-Based Inference for Real-World Population Estimation

Project Proposal
BSc Statistical Data Science, Year 3

November 5, 2026

Abstract

The German Tank Problem asks for the size N of a population labelled $1, \ldots, N$ given a random sample of k labels drawn without replacement. The classical solution (the UMVUE $\hat{N} = m(1 + 1/k) - 1$ with m the sample maximum) is frequentist and assumes a textbook sampling mechanism. This project develops a fully Bayesian treatment: (i) exact posterior results under a family of informative priors, (ii) extensions to realistic violations of the textbook assumptions (multiple factories, selection bias, gaps and batch structure, time-varying production), and (iii) an amortised neural posterior estimator for settings where the likelihood is awkward or intractable, benchmarked against the exact posterior where one exists. Applications include software defect counting, order-volume estimation, wildlife tagging, counterfeit detection and NFT mint-size estimation.

1 Background and motivation

Let $S = \{s_1, \ldots, s_k\}$ be drawn uniformly without replacement from $\{1, \ldots, N\}$ and let $m = \max S$. The likelihood depends on the data only through (m, k):

$$\mathbb{P}(S \mid N) = \binom{N}{k}^{-1} \mathbf{1}\{N \geq m\}, \qquad \mathbb{P}(M = m \mid N) = \binom{m-1}{k-1}\binom{N}{k}^{-1}. \tag{1}$$

Since m is sufficient, all inference reduces to a one-dimensional discrete posterior on $n \geq m$.

Starting point. This proposal builds on an MSc dissertation on the topic (Gontla, QMUL, 2021), which derives the frequentist estimator and its variance $\mathrm{Var}(\hat{N}) = \frac{(N+1)(N-k)}{k(k+2)}$, and the Bayesian posterior under a flat prior, with mean $\frac{k-1}{k-2}(m-1)$, and compares them by simulation ($N = 1000$, $k = 20$). That work also attempts a Poisson prior. On reading it, several points offer a clear opportunity for a rigorous follow-up:

- The Poisson-prior experiment sets the rate μ to the mean of the grid $\{0, \ldots, 2000\}$ and calls this an MLE; this is not a data-driven estimate. A principled approach is to place a hyperprior on μ (or use empirical Bayes / marginal likelihood).
- Sensitivity of the posterior to the prior is reported as an "error". In Bayesian analysis this is an expected feature that should be *quantified* (prior-sensitivity analysis, robust Bayes), not removed.
- Methods are compared via the mean and standard deviation of point estimates, which cannot assess uncertainty quantification. Proper scoring rules and interval coverage are needed.

- The estimators are only tested under the idealised sampling mechanism.

2 Research questions

RQ1. How do informative priors (truncated/geometric, Poisson, negative binomial, log-uniform) change the posterior of N, credible intervals and frequentist operating characteristics, and how sensitive are conclusions to the prior?

RQ2. How can the model be extended to multiple factories/serial series, selection bias, gaps, and time-varying production, and what is the cost of ignoring these features?

RQ3. Can an amortised neural posterior estimator recover the exact posterior in the base case and give calibrated inference in extended settings with intractable likelihoods?

RQ4. How do these methods perform on realistic applied datasets?

3 Mathematical framework

3.1 Posterior under a general prior

For a prior $\pi(n)$ on $\mathbb{N}$,

$$\pi(N = n \mid m, k) = \frac{\binom{n}{k}^{-1}\pi(n)}{\sum_{v \geq m}\binom{v}{k}^{-1}\pi(v)}, \qquad n \geq m. \tag{2}$$

Improper flat prior $\pi(n) \propto 1$. Using $\sum_{v \geq n}\binom{v}{k}^{-1} = \frac{k}{k-1}\binom{n-1}{k-1}^{-1}$, the posterior is proper for $k \geq 2$ and

$$\pi(N = n \mid m, k) = \frac{k-1}{k}\binom{m-1}{k-1}\binom{n}{k}^{-1}, \qquad \mathbb{P}(N \geq n \mid m, k) = \binom{m-1}{k-1}\Big/\binom{n-1}{k-1}. \tag{3}$$

The posterior mean is $\frac{(k-1)(m-1)}{k-2}$ (finite for $k \geq 3$), and the variance is finite for $k \geq 4$. The closed-form survival function gives exact one-sided credible bounds, and for large m the tail behaves like $(m/n)^{k-1}$, so the upper $(1-\alpha)$ bound is approximately $m\,\alpha^{-1/(k-1)}$. The HPD interval is always of the form $[m, u_\alpha]$ because the posterior mass function is decreasing in n for $n \geq m$ under a flat prior. This is the first deliverable: a verified derivation and a numerical check.

Informative priors. We study

- *Truncated uniform* on $\{1, \ldots, \Omega\}$ (capacity constraint), with sensitivity in Ω;
- *Geometric/exponential decay* $\pi(n) \propto \lambda^n$;
- *Poisson and negative binomial* priors centred on expected production capacity, with hyperpriors on the rate (a Gamma hyperprior makes the marginal a negative binomial);
- *Log-uniform / scale-invariant* $\pi(n) \propto 1/n$, a standard weakly informative choice for scale parameters.

Posterior summaries (mean, median, MAP, HPD interval) are computed by exact summation over a truncated support with a tail-error bound.

3.2 Realistic extensions

Multiple factories / sub-series. Let there be F series with sizes $N_1, \dots, N_F$ and observed tagged items (f_i, s_i). If the k items are a uniform random subset of the pooled population, with k_f items seen from series f and maxima m_f,

$$\mathbb{P}(S \mid N_{1:F}) = \binom{\sum_f N_f}{k}^{-1} \prod_f \mathbf{1}\{N_f \geq m_f\}. \tag{4}$$

Priors are placed on N_f hierarchically, $N_f \sim \text{NegBin}(r, p)$ with shared hyperparameters, and on F itself (e.g. a Dirichlet-process or mixture-of-finite-mixtures prior). Posterior inference is by MCMC (Stan/NumPyro or a custom Gibbs sampler on the discrete N_f).

Selection bias. Suppose item s is observed with weight $w(s) > 0$ (e.g. older units fail more often). The likelihood for a single draw is $\mathbb{P}(s \mid N) = w(s)\mathbf{1}\{s \leq N\}/Z_N$ with $Z_N = \sum_{j \leq N} w(j)$. The proportionality $P(s \mid N) \propto w(s)\mathbf{1}_{s \leq N}$ must therefore keep the *N-dependent normaliser* Z_N; omitting it removes the penalty for large N. For sampling without replacement the exact likelihood is a Wallenius-type (sequential) form, which we will approximate or evaluate by simulation. Parametric families such as $w(s; \theta) = s^{-\theta}$ or $\exp(-\theta s)$ are studied with θ given a prior, so selection strength is learned rather than assumed.

Gaps and batch structure. Serial numbers may skip values (e.g. rejected units, check digits, lot blocks). We model the observed labels as a thinned or blocked sequence, $s = g(j)$ with $j \leq N$ a latent unit index, and treat the thinning probability or block size as parameters with priors.

Time-varying production. For production rate λ_t per month and cumulative total $N_t = \sum_{u \leq t} \lambda_u$, we combine the tank-style serial-number data with a state-space prior over λ_t (random walk on $\log \lambda_t$), giving a posterior over production *rates* rather than a single N.

3.3 Simulation-based inference with deep learning

When extended models have intractable or expensive likelihoods, we train a neural posterior estimator $q_\phi(N \mid S)$ on simulated pairs $(N, S) \sim \pi(N)\, p(S \mid N)$:

- **Set encoder:** a permutation-invariant DeepSets / Set Transformer embedding of the sampled serial numbers (plus prefix/series features);
- **Density head:** a categorical head over $\{1, \dots, N_{\max}\}$ (or a mixture density / normalising flow for continuous relaxations), trained by minimising the forward KL (negative log-likelihood);
- **Validation:** in the base case, compare q_ϕ to the exact posterior in total variation and KL; in extended cases, use simulation-based calibration (SBC) and coverage checks.

This gives a concrete, checkable test of whether amortised inference is trustworthy, since the exact answer is known in the baseline.

4 Real-world applications

At least two applications will be demonstrated on publicly available or synthetic-but-realistic data (e.g. public bug trackers, on-chain token metadata), plus the historical WWII data (statistical estimates of 169, 244 and 327 versus German records of 122, 271 and 342 for June 1940, June 1941 and August 1942) as a reference benchmark.

Domain	Quantity N	Main violation to model
Software defects	Total bugs in a release, from sequential issue/patch IDs found in testing	Non-random discovery; IDs shared with non-bug tickets (gaps)
E-commerce	Orders or sign-ups in a period from leaked invoice/transaction IDs	Time-varying rate; multiple ID series
Wildlife tagging	Population size from sequentially numbered tags	Capture heterogeneity; tag loss
Supply chain	Production batch size; detection of serials beyond legal limits (counterfeits)	Multiple factories; selection toward visible units
NFT / token mints	Collection size from sampled on-chain token IDs	Sampling via marketplaces biased to popular IDs

5 Methodology and evaluation

1. **Exact baseline.** Implement the exact posterior with numerically stable log-binomials; verify against Monte Carlo.
2. **Prior study.** Compare priors and hyperpriors on bias, RMSE, *log score*, *CRPS*, interval score, and coverage of 95% credible intervals across (N, k) grids (e.g. $N \in \{100, 1000, 10^5\}$, $k \in \{4, 10, 20, 100\}$), including misspecified-prior cases.
3. **Frequentist comparison.** Compare against the UMVUE $m(1 + 1/k) - 1$, the MLE m, and confidence intervals derived from the distribution of m/N.
4. **Extended models.** Fit the multi-factory, selection-bias and gap models; run *misspecification experiments* (data from the extended model, fit with the naive model) to quantify the damage.
5. **Neural estimator.** Train and validate q_ϕ as above; report calibration and amortisation speed-up.
6. **Applications.** Apply the models, report posteriors and decision-relevant probabilities, e.g. $\mathbb{P}(N > N_{\text{legal}} \mid S)$ for counterfeit detection.

Implementation: Python (NumPy/SciPy, PyTorch, NumPyro/PyMC) with R for comparison to the existing code; all experiments seeded and version-controlled.

6 Expected deliverables

- A written report with verified derivations (flat-prior closed forms, exact credible bounds, hierarchical extensions).
- An open, reproducible code repository and notebooks.
- A systematic comparison of priors and estimators using proper scoring rules and coverage.
- A neural posterior estimator with calibration diagnostics.
- Applied case studies and a decision-oriented discussion of uncertainty.

Weeks	Tasks
1–2	Literature review; derive and verify exact posterior; implement baseline in Python/R
3–4	Informative priors and hyperpriors; prior-sensitivity analysis; scoring-rule framework
5–6	Multi-factory and gap models; MCMC implementation; misspecification experiments
7–8	Selection-bias model and time-varying production model
9–10	Neural posterior estimation; SBC and comparison with exact posterior
11	Application case studies
12	Final report, code clean-up, presentation

7 Timeline (12 weeks)

8 Risks and mitigation

- **Identifiability:** selection-strength and population size may be confounded with small k; mitigate with informative priors and prior predictive checks.
- **Computation:** discrete-parameter MCMC may mix poorly; mitigate by marginalising N_f analytically where possible.
- **Data availability:** if real datasets are limited, use realistic simulators calibrated to published figures.
- **Scope:** extensions are ordered by priority (RQ1 and RQ2 first; time-varying model is optional).

References

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